

Aspect-Level Sentiment Decline and Return Is Not Estimable from Two Large Public Review Corpora: A Negative Feasibility Study with a Density Bound

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Abstract

Review-analytics work routinely models how sentiment toward a product attribute evolves after a negative event and reads the return of sentiment as recovery. We asked whether that construct is estimable at all from public review corpora. Operational definitions of shock, recovery and censoring were fixed in a version-controlled protocol before any data were inspected, then applied to Amazon Reviews 2023 and MobileRec. At the product level, 313 of 11,925 product-months met a 30-mention threshold (2.6 %) and no shock was detected; a denser extract yielded five analysable episodes against the 65 events needed to detect a hazard ratio of 2.0 at 80 % power. Aggregating to category level restored whole-review density (955 qualifying category-months) but not identifiability at the aspect level, where the series do not form at all. At category level, where whole-review series do exist, the picture is mixed and we report it as such: a threshold rule found 23 shocks against 28.4 ± 3.8 expected under a circular block bootstrap null at 999 replicates, a ratio of 0.81 with empirical $p = 0.95$, while PELT applied to the same series found six change points against 1.77 ± 1.35 expected, a ratio of 3.40 at $p = 0.015$ uncorrected, or $p = 0.060$ under Bonferroni across the four tests performed. A positive control shows why the first result is weak evidence of anything: the threshold criterion fires on every clean synthetic series and recovers only two-thirds of injected two-standard-deviation shocks. We therefore claim neither presence nor absence of structure at category level, since our criterion cannot distinguish them; the claim we do make is narrower, that the aspect-level series required to study the question do not exist. An unvalidated lexicon measure, reported for robustness only, likewise fails to reject, though it differs from the star rating on outcome composition and censoring and should not be described as agreeing with it. Returning to the aspect level, none of 7,084 aspect cells reached the threshold, at a median of one mention per cell. We derive a density bound: the number of aspects that can reach the threshold is at most Rm/T , for R reviews per unit-period, m aspects named per review and threshold T . At $T = 30$ the bound caps MobileRec at $1.16m$ aspects and Amazon at $0.23m$. We did not measure m , so the cap is conditional on it, and we report the bound's sensitivity to both m and T rather than asserting a single number. Repeat reviewing of the same item runs at 0.159 % and 0.063 % in the two corpora, so recovery of an average cannot be separated from turnover in who is reviewing. The obstacle is the quantity of text, not the choice of method.

Data and code. Every figure, table and number in this article is reproduced by the scripts and protocol archived at

<https://github.com/ssvakil/aspect-recoverability>

The archive contains the operational definitions fixed before data inspection, the analysis code with its self-test suite, and the figure generation script.

Keywords: online reviews; sentiment dynamics; negative results; survival analysis; construct validity; reproducibility

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1 Introduction

Online reviews are routinely mined for signals about how customer perception changes over time. Aspect extraction from review text dates to Hu and Liu (2004) and remains active (Zhou et al., 2023). A recurring pattern in the applied literature is to construct an attribute-level sentiment series, identify a decline, observe whether sentiment later returns, and interpret the return as recovery (Park and Kim, 2024; Roelen-Blasberg et al., 2023; Yang et al., 2023). The managerial appeal is obvious: if some attributes recover and others do not, firms could prioritise repairs accordingly.

Our study began as an attempt to formalise this idea as a time-to-event problem. We intended to model the time from an attribute-level sentiment shock to recovery, treating episodes that had not recovered by the end of the horizon as administratively censored rather than as a distinct event type, and to test whether non-recovering shocks predicted subsequent customer disengagement. Before committing to that analysis we did what is rarely reported: we asked whether the data could support it.

It cannot. This paper reports why, and we argue the reasons generalise beyond our particular design.

Our contribution is threefold. First, we provide a quantified feasibility assessment of attribute-level recovery estimation on two corpora that are standard in this literature, showing that the required longitudinal density is absent by one to two orders of magnitude. Second, we identify a construct validity problem that applies to any population-level recovery claim: because repeat reviewing is rare, recovery of an average is not separable from turnover in who is averaged. Third, we show that the choice of null control matters materially, and that the permutation null most readily available is biased toward declaring detection successful.

Negative feasibility results are underreported, which means the same infeasible designs are attempted repeatedly. Publication bias toward positive findings is long documented (Rosenthal, 1979; Easterbrook et al., 1991; Fanelli, 2010), operates through non-writeup as well as non-acceptance (Franco et al., 2014), and motivates registered reports (Chambers, 2013) and broader open-research norms (Nosek et al., 2015; Ioannidis, 2005). Feasibility failures are a costly omission in particular, because the cost of rediscovering them falls on every group that attempts the design. We therefore report our thresholds, our pre-specified definitions, and our failure points in full.

2 Related work

2.1 Attribute-level sentiment over time

That customer evaluations of specific attributes shift over the consumption cycle is long established. Mittal et al. (1999) analysed longitudinal data from 5,206 automobile owners and showed that attribute-level performance, satisfaction, and behavioural intentions must be examined intertemporally because the relationships among them change as consumption proceeds, with attribute weights shifting according to their salience over time. Later computational work reaches compatible conclusions with different methods (Park and Kim, 2024; Roelen-Blasberg et al., 2023). Adjacent work in requirements engineering classifies app reviews into actionable categories (Maalej and Nabil, 2015), the closest analogue to attribute extraction in the mobile setting.

Xia et al. (2021) model the dynamic evolution of user preferences and item features at the aspect level, detecting both gradual drift and abrupt change points across three real-world datasets. Their work establishes that aspect-level series contain detectable structure; our results are not in conflict with theirs, since we address a narrower question about a specific event-based construct.

2.2 Recovery after failure

The marketing literature has examined recovery for decades. The service recovery paradox holds that a well-handled failure can leave a customer more satisfied than if nothing had gone wrong. The meta-analysis by de Matos et al. (2007) found support for this only in post-recovery satisfaction, with an effect size of 0.125, and no significant effect on repurchase intentions — that is, recovered satisfaction does

not ensure loyalty. They further report that study design, subject type, and service category moderate the effect, and characterise the underlying body of findings as inconsistent.

Two features of that literature are relevant here. It measures stated constructs in scenario settings rather than observed behaviour, and its unit of analysis is a failure episode involving a firm response. Our design observes behaviour but cannot observe the firm response, which is why we framed the construct as perceptual rather than managerial.

Negativity bias is a standing explanation for why declines and recoveries need not be symmetric (Baumeister et al., 2001; Rozin and Royzman, 2001): negative information is weighted more heavily and is more resistant to disconfirmation than positive information of comparable magnitude. Asymmetry between satisfaction and switching is also long documented: LaBarbera and Mazursky (1983) found that repurchase of a brand was influenced by lagged intention, whereas switching behaviour was more sensitive to dissatisfaction with brand consumption.

2.3 How this study differs

Table 1 positions the present study against the work closest to it on the dimensions that decide whether recovery is estimable. The columns are not measures of quality: the studies listed answer questions of their own, and most had no reason to report what we look for here. They are the properties a recovery analysis needs.

Three patterns emerge. Work that observes real behaviour at the attribute level is cross-sectional or models gradual drift, not discrete events. Work that treats failure and recovery as events comes from service marketing and measures stated intention in scenario settings rather than observed behaviour. And no study in either group reports whether its data were dense enough to support the analysis attempted, which is the question this paper is about.

Table 1. The present study against the closest prior work. **Obs.:** observational data rather than survey or scenario. **Long.:** repeated measurement over time. **Event:** failure and recovery treated as discrete events rather than as gradual drift. **TTE:** a time-to-event model. **Null:** a resampling null against which detections are compared. **Density:** feasibility of the data for the analysis attempted is reported. Characterisations follow each study’s published account of its own design.

Study	Unit	Obs.	Long.	Event	TTE	Null	Density
<i>Attribute extraction and ranking from reviews</i>							
Hu and Liu (2004)	aspect	yes	no	no	no	no	no
Maalej and Nabil (2015)	review	yes	no	no	no	no	no
Zhou et al. (2023)	feature	yes	no	no	no	no	no
Roelen-Blasberg et al. (2023)	attribute	yes	no	no	no	no	no
Yang et al. (2023)	attribute	yes	no	no	no	no	no
Park and Kim (2024)	feature	yes	no	no	no	no	no
<i>Temporal dynamics of attribute sentiment</i>							
Xia et al. (2021)	user $\times$ item $\times$ aspect	yes	yes	no	no	no	no
Mittal et al. (1999)	attribute	no	yes	no	no	no	n/a
<i>Failure and recovery</i>							
LaBarbera and Mazursky (1983)	overall	no	yes	no	no	no	n/a
de Matos et al. (2007)	episode	no	mix	yes	no	no	n/a
<i>This study</i>							
Present work	aspect $\times$ unit $\times$ month	yes	yes	yes	attempted	yes	yes

The final row reads "attempted" rather than "yes" deliberately. We specified a time-to-event analysis and could not run it, and the reason is the subject of this paper. Marking it "yes" would claim a contribution we did not make.

The comparison also shows why the gap persisted. A study needs observational data for behaviour, longitudinal structure for dynamics, and an event framing for recovery. The first two are common; the third belongs to a literature that works with surveys and scenarios. Combining all three is what exposes the density constraint, and it is only visible once the combination is attempted.

2.4 What has not been done

We found no prior work that models attribute-level recovery as a time-to-event outcome on naturally occurring longitudinal review data, or that separates recovery of the original reviewer cohort from recovery of the population average. We state this carefully: our search was targeted rather than systematic, so it establishes that we did not find such work, not that none exists (Section 7). That apparent absence motivated the study. Our results suggest one reason for it.

3 Method

3.1 What the construct is, and is not

A clarification that bears on every result below. We observe no intervention dates, no product releases, no recalls and no firm responses. A "shock" here is a sustained fall in measured sentiment, identified from the series itself, and "recovery" is a return of that series toward its earlier level.

This is therefore a study of *sentiment decline and return*, not of service recovery in the sense of de Matos et al. (2007), where a failure event and a firm response are both observed and the outcome is satisfaction or stated intention. The two constructs share a shape and little else. Where we use the word recovery we mean the former, and no claim about the latter follows from our results.

The distinction also limits the interpretation available even had detection succeeded. A series that falls and returns is consistent with a defect introduced and repaired, with a change in who is reviewing, with a seasonal effect, and with a shift in platform ranking. Without intervention dates these are not separable, which is why Section 6.3 lists documented intervention timestamps among the data properties a usable study would need.

3.2 Pre-specified definitions

All definitions in Table 2 were fixed before any data were inspected and are archived in the repository. No threshold or classification rule was changed after data inspection. One deviation of scope was made and is recorded in the deviation log: after the product- and app-level analyses proved infeasible, the unit of analysis was moved to the app category. That analysis was not covered by the protocol and is reported as exploratory throughout.

Table 2 also lists the disengagement definition, which belonged to the planned downstream analysis. That analysis was never reached, and no disengagement result is reported here.

The censoring rule deserves emphasis. A shock occurring late in the observation window can never be observed to recover. Classifying such episodes as non-recovered rather than censored inflates estimated irreversibility systematically, and the size of the inflation grows with the recovery horizon.

Table 2. Operational definitions, fixed before data inspection.

Construct	Definition
Qualifying month	At least 30 mentions in that unit-month; months below the threshold are dropped and never interpolated
Baseline	Mean sentiment over the six months preceding shock onset
Negative shock	Drop of at least 1 SD below baseline, sustained for at least two consecutive months
Recovery	Return to at least 90 % of baseline, sustained for at least three consecutive months
Non-recovery	No recovery within 18 months, <i>with the full 18 months observed</i>
Right-censored	Fewer than 18 months of post-shock data available
Disengagement	No further review of the brand for 12 months, conditional on the user reviewing elsewhere in that window

3.3 Detection procedure

Monthly sentiment series were constructed per unit, filtered to qualifying months, and split at gaps so that no shock could be detected across a discontinuity. Shocks were located by comparison against a trailing baseline window, then classified as recovered, non-recovered, or censored according to Table 2. The protocol named the PELT algorithm (Killick et al., 2012), available as described by Killick and Eckley (2014), as a robustness check on shock localisation, with Bayesian online change point detection (Adams and MacKay, 2007) as a sequential alternative and the formulations of Barry and Hartigan (1993) and Fearnhead and Liu (2007) as further options. None of these was run. The feasibility constraints reported in Section 5 removed the need: a robustness check on the location of shocks is only meaningful once shocks are distinguishable from noise, which they were not. All reported counts therefore use the threshold rule, which is transparent and tied directly to the pre-specified definitions.

3.4 Null controls

Detection procedures of this kind will find events in pure noise, so an explicit null is required. We report two.

The obvious null is a permutation of the series. This is biased. Real sentiment series are autocorrelated, so adjacent months resemble one another and the local standard deviation of a trailing window is small. Permutation destroys that autocorrelation and inflates the local standard deviation, which raises the detection threshold and mechanically suppresses shock counts. A comparison against a permutation null therefore favours the hypothesis being tested.

We instead use a circular block bootstrap (Politis and Romano, 1992), a variant of the moving block bootstrap (Künsch, 1989) that wraps the series on a circle to remove edge effects, and which preserves short-range autocorrelation while randomising where episodes fall in time. The stationary bootstrap (Politis and Romano, 1994) would serve equally; we fix the block length rather than drawing it geometrically, and note that automatic block-length selection (Politis and White, 2004; Patton et al., 2009) would be preferable in a confirmatory study. Our implementation includes a self-test demonstrating, on a synthetic AR(1) series containing no injected shocks, that the permutation null returns fewer shocks than the block bootstrap null.

Blocks are of fixed length six months, set equal to the baseline window so that within-baseline dependence is preserved inside a single block; the length was not tuned. We draw 999 replicates, giving a Monte Carlo standard error of the null mean of $3.77/\sqrt{999} = 0.12$ shocks against a null standard deviation of 3.77. All p -values are empirical quantiles of the bootstrap distribution under the $(b+1)/(B+1)$ convention, which prevents a p of exactly zero. Block length was varied over 3, 6 and 12 months as a check: the observed-to-null ratio moves only from 0.78 to 0.85 under star ratings, so the conclusion does not rest on the value chosen.

3.5 Power

The planned analysis was a Cox proportional hazards model (Cox, 1972), preceded by Kaplan–Meier estimation of time to recovery (Kaplan and Meier, 1958). Schoenfeld’s formula (Schoenfeld, 1983) gives the number of *events*, not observations, required to detect a hazard ratio:

$$d = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2}{p(1-p)(\ln HR)^2}, \quad (1)$$

where p is the exposed fraction. At $\alpha = 0.05$, 80 % power, and balanced exposure, detecting $HR = 2.0$ requires 65 events and $HR = 1.5$ requires 191. Because censored episodes contribute person-time but not events, the number of episodes needed exceeds these figures.

We also report events per parameter (EPP), since Schoenfeld’s formula assumes a single covariate of interest and the planned model has more. The conventional floor of ten events per parameter derives from Peduzzi et al. (1996); it has been argued relaxable in some circumstances (Vittinghoff and McCulloch, 2007) and challenged as lacking a general rationale (van Smeden et al., 2016). We report EPP at three and six parameters so the reader may apply whichever standard they prefer.

3.6 Immortal time in the recovery covariate

Recovery status is determined over an 18-month window, which makes it a time-dependent quantity. Entering it into a Cox model as a baseline covariate would introduce immortal time bias: an episode cannot be labelled recovered unless it survived long enough for recovery to be observed, so the recovered group would contain disproportionately late-failing observations by construction, and the model would report a protective effect of recovery that is an artefact of the definition. The bias was first identified in cohort studies of transplantation survival and has since been documented across many pharmacoepidemiological studies (Suissa, 2007, 2008; Lévesque et al., 2010); its structure and remedies are reviewed by Platt et al. (2019) and Yadav and Lewis (2021). We therefore specified recovery as a time-varying covariate, with person-time accruing in the non-recovered state from shock onset and switching at the month recovery is confirmed. Landmark analysis (Mi et al., 2016) and the alternatives compared by Karim et al. (2016) would be acceptable substitutes; the time-varying specification was fixed in the protocol.

3.7 Reproducibility

The analysis code contains 77 self-tests covering shock classification, censoring classification, power arithmetic, series segmentation, null construction, and recovery attribution. Tests assert behaviour on series whose correct classification is known by construction; for instance, that a truncated permanent decline is labelled censored rather than non-recovered. These establish that the classification logic is correct given a detection. They do not establish that detection occurs, which is a separate property and is measured by the positive control of Section 5.6. A further 78 cover the data loaders and the figure generation script, giving 155 in total. All code, definitions, and test suites are at <https://github.com/ssvakil/aspect-recoverability>.

4 Data

4.1 Scope of the empirical claims

We analysed a 500,000-interaction scan of MobileRec, about 2.6 % of its 19.3 million interactions, and an 82,041-review extract covering the 200 densest products of one Amazon category. Neither is the full corpus, and the Amazon extract is deliberately non-representative: it selects dense products because density is the resource the analysis consumes.

Our empirical statements are therefore about these subsamples at these thresholds, and we have tried to phrase them that way throughout. The density bound is the exception: it is arithmetic on R , m and T ,

and applies to whatever panel a study actually assembles. A larger scan would raise the absolute counts but would not raise R , since R is a rate; only reviewing behaviour or a different sampling design would do that.

4.2 Amazon Reviews 2023

We used the public research release covering 571.54 million reviews from May 1996 to September 2023 (Hou et al., 2026). Random sampling is inappropriate here: the analysis consumes longitudinal density, and a random sample of reviews spreads thinly across millions of products so that no product reaches the mention threshold. We therefore selected the densest products and retained all of their reviews. This trades representativeness for density, and findings accordingly generalise to heavily-reviewed products only.

4.3 MobileRec

MobileRec contains 19.3 million interactions across 10,173 apps from 0.7 million users, ordered chronologically and filtered to a 5-core (Maqbool et al., 2023). The 5-core criterion means each user reviewed five *distinct* apps. This is not the same as reviewing one app repeatedly, and only the latter supports cohort-level recovery analysis — a distinction that is easy to miss and that proved decisive.

5 Results

5.1 Density is insufficient at the product level

Table 3 summarises the feasibility of each configuration attempted. In Amazon Reviews 2023, the 200 densest products of a category yielded 82,041 reviews spanning 2005–2023, of which 313 of 11,925 product-months (2.6 %) met the 30-mention threshold. Zero shocks were detected. Repeating the analysis on the full extract rather than a subsample produced the same result, confirming that this was not an artefact of sampling.

Table 3. Feasibility across configurations. Qualifying unit-months are those meeting the 30-mention threshold. Scan size and observation period are stated for every row because they differ, and shares computed on different denominators are not directly comparable. Rows within a block share a basis. Balanced-panel denominators were absent from an earlier version of this table and have since been logged by re-running the analysis; they are the category-months actually present in each panel, which falls below the product of categories and months because sparse months are dropped.

Configuration	Scan	Units	Unit-months	Qualifying	Share
<i>Full observation period</i>					
Amazon, product (All_Beauty)	82,041	200	11,925	313	2.6 %
MobileRec, app	300,000	n.r.	128,251	271	0.2 %
MobileRec, app	500,000	8,603	170,846	827	0.5 %
MobileRec, category	500,000	48	5,134	2,286	44.5 %
<i>Window 2018-01 to 2021-12</i>					
MobileRec, category, unbalanced	500,000	48	2,299	1,788	77.8 %
presence ≥ 50 %	500,000	48	2,291	1,589	69.4 %
presence ≥ 75 %	500,000	46	2,150	955	44.4 %
presence ≥ 90 %	500,000	10	453	0	0.0 %

Figure 1 shows the progression across the three full-period configurations, which share a basis and are therefore comparable. MobileRec at the app level is sparser than Amazon at the product level, at 0.5 % of app-months qualifying against 2.6 %; a smaller 300,000-interaction scan of the same corpus gives 0.2 %. Neither is close to usable.

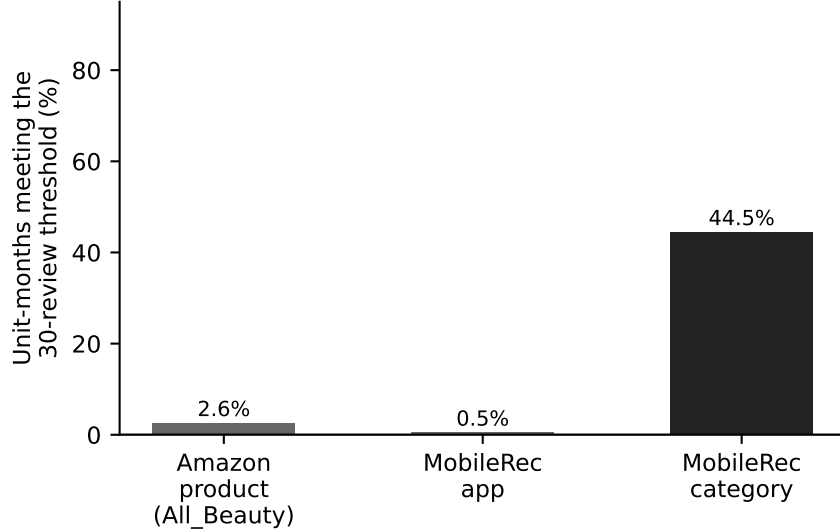


Figure 1. Share of unit-months meeting the 30-mention threshold at each level of aggregation, all over the full observation period on a 500,000-interaction scan for MobileRec. The category figure here, 44.5 % (2,286 of 5,134), is *not* the 44.4 % quoted elsewhere: that is the windowed balanced panel, 955 qualifying category-months out of a denominator the run did not log. It lies below the $46 \times 48 = 2,208$ maximum, since sparse months are dropped, but we do not state it, because recovering it by dividing 955 by the reported share would be a derived figure presented as a measured one. The two percentages are near-identical by coincidence and rest on different bases; we flag this because the resemblance misled us during analysis.

5.2 Statistical power is unattainable at the product level

A second Amazon extract, from a category with heavier reviewing than All_Beauty, produced five analysable episodes. Against the 65 events required for $HR = 2.0$, this is short by a factor of thirteen; against the 191 required for $HR = 1.5$, by a factor of thirty-eight. Figure 2 places all three configurations against both requirements.

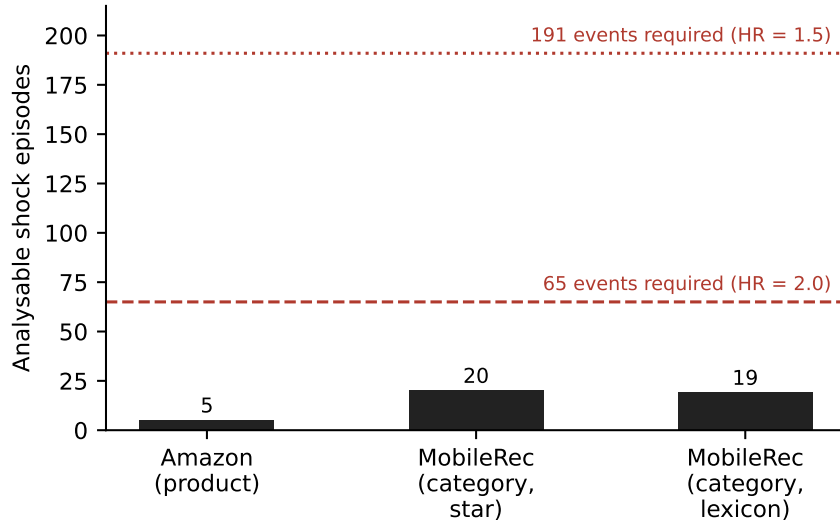


Figure 2. Analysable shock episodes obtained, against the events required by Schoenfeld's formula at 80 % power: five for Amazon at product level, 20 for the category panel under star ratings and 19 under lexicon polarity. No configuration approaches either threshold.

Inverting Schoenfeld's formula gives the smallest effect the obtained event count could detect. With

the 20 analysable episodes from the star-rating measure, the minimum detectable hazard ratio at 80 % power is 3.50 (Figure 3). Hazard ratios of that magnitude are not plausible for the effect under study, so the design is not merely underpowered but incapable of addressing the question at this sample size.

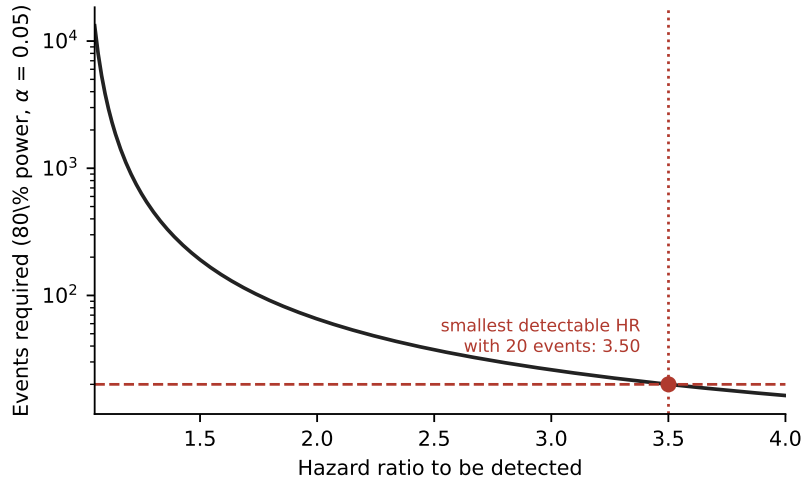


Figure 3. Events required as a function of the hazard ratio to be detected, with the largest obtained episode count marked. The intersection gives the minimum detectable effect.

5.3 Cohort recovery is not computable

In a 300,000-interaction sample of MobileRec, 188 of 299,812 user–app pairs contained more than one review, a repeat rate of 0.063 %; within the balanced panel it was 0.059 %. In the Amazon extract, 130 of 81,902 user–product pairs repeated, a rate of 0.159 %. Amazon reviewers return to the same item about two and a half times as often as MobileRec users (Figure 4).

Neither figure supports a cohort analysis, and the shortfall is not marginal. With 23 shock episodes and a 0.159 % repeat rate, the expected number of pre-shock reviewers who return after their shock is below one. We set a working convention of 2 % in our code as the point below which the cohort estimate is withheld rather than reported; that convention is ours and we know of no standard for it, but the observed rates are more than an order of magnitude below any threshold one might reasonably choose, so nothing in our conclusion turns on where it is set. Measuring both matters: one corpus would leave open that the rate reflects a single platform’s design, whereas two corpora with different sampling schemes and different product types point to the behaviour rather than the dataset.

These rates are for the same user reviewing the same *item*. Our shocks, however, are defined at category level, so the item rate is not the relevant one, and reporting only it understated what the data hold. Measuring the coarser units changes the picture materially (Table 4).

Table 4. Repeat-observation rates in the balanced MobileRec panel at four units. A cohort analysis needs repeat observation at the unit the shock is defined on, which here is the category.

Unit	Pairs	With repeats	Rate
User × item	74,736	44	0.059 %
User × category	73,563	1,205	1.638 %
User × aspect	14,244	198	1.390 %
User × category × aspect	14,440	5	0.035 %

The user-category rate is 1.64 %, twenty-eight times the item rate. MobileRec’s 5-core filter guarantees five distinct apps per user, and those apps may fall in one category, so a user contributes repeated observations of a category without ever reviewing an app twice. An earlier version of this paper described

the repeat rates as two orders of magnitude below what a cohort analysis needs. At category level that is not so: the shortfall is a factor of a few, not a hundred, and the honest statement is that a category-level cohort analysis is marginal rather than impossible.

We did not run one, and on reflection "marginal" overstates what these rates allow. With 23 shock episodes at a 1.64 % rate, the expected number of returning pre-shock reviewers across the entire study is $23 \times 0.0164 = 0.38$: fewer than one person. A tenfold larger panel would yield about four. The finest unit, user by category by aspect, collapses to 0.035 %. The accurate statement is that a category-level cohort analysis is out of reach at this scale, by a wide margin, though the margin is a factor of tens rather than the factor of hundreds the item-level rate suggested. Which unit one quotes changes the size of the gap but not the verdict.

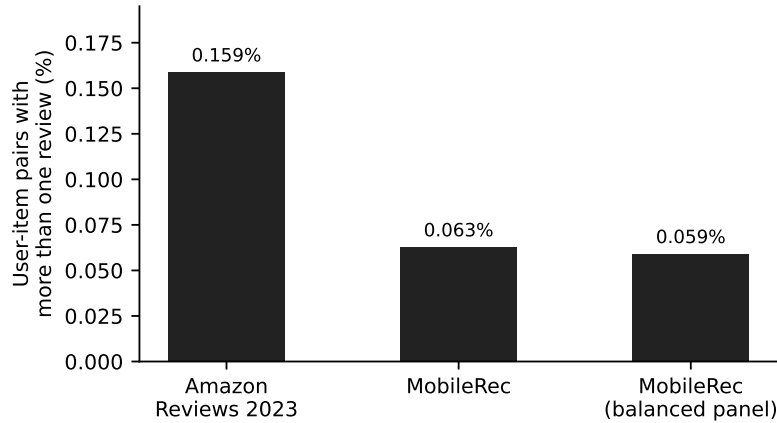


Figure 4. Share of user–item pairs carrying more than one review. Neither corpus approaches the level a cohort analysis requires.

This forecloses the analysis that motivated the study. If sentiment toward a unit returns after a shock, that return may be because the reviewers who complained changed their assessment, or because they left and different people arrived. These have opposite managerial implications and are empirically indistinguishable without repeat observations of the same individual. Any recovery reported on such data is population-level, and the distinction is not a refinement but a limit on what the measurement can mean.

5.4 Aggregation restores density but not identifiability

Aggregating to app category raised the qualifying share to 77.8 %. That figure recomputes category membership each month, so it may reflect apps entering and leaving rather than sentiment change. We therefore constructed balanced panels retaining only apps present across a minimum share of the window (Figure 5).

At 75 % presence, 657 apps and 46 categories survived, with 955 category-months qualifying out of 2,150 category-months present in that panel, a share of 44.4 % (95 % CI [42.3, 46.5], Wilson). The denominator is now logged. An earlier version of this paper quoted the share without it and, worse, once recovered a denominator by dividing the count by the share and reported the result as measured; the logged figure is 2,150 and the derived one was 2,151. Compared against the 77.8 % obtained without balancing over the same window, this is a relative reduction of 43 %: composition change contributes materially to the apparent density, and roughly two of every five qualifying category-months in the unbalanced figure depend on apps that are not present throughout. Density survives balancing, but not untouched, and the unbalanced figure should not be quoted as though it described a stable panel. At 90 % presence only 10 apps remained and no category-month qualified at all, which sets a firm limit on how far the balanced-panel result can be pushed.

Shock detection requires 26 consecutive qualifying months (six baseline, two shock, eighteen horizon). Sixteen categories met this criterion, with the longest runs reaching 43 months (Figure 6).

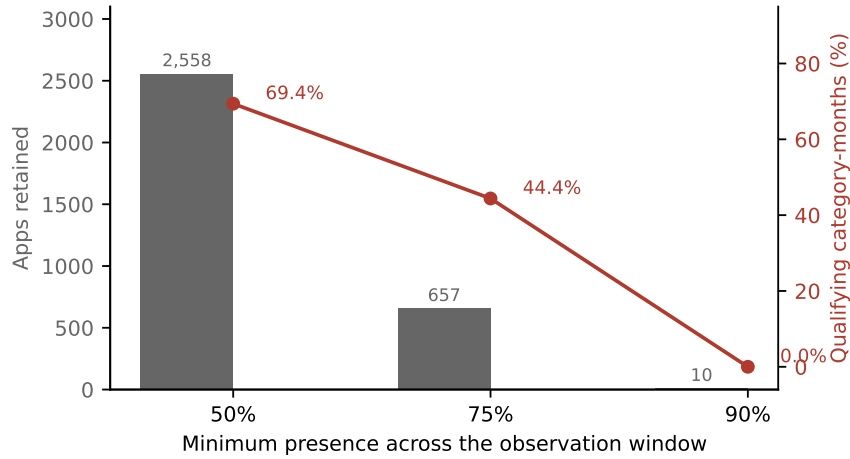


Figure 5. Apps retained and qualifying category-months as the presence requirement tightens. Density survives at 75 % presence but collapses at 90 %, indicating that composition is not fully stable even in the balanced panel.

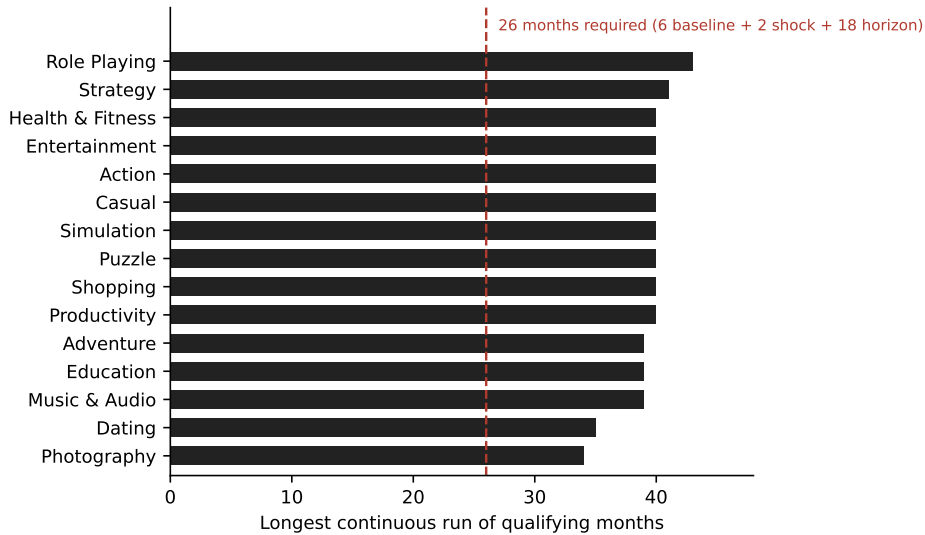


Figure 6. Longest continuous run of qualifying months per category under the 75 % presence panel, against the 26 months a full shock-and-recovery window requires. Top fifteen categories shown.

5.5 Detected shocks are indistinguishable from noise

Two counts must be distinguished here. Sixteen categories carry a run of 26 or more contiguous qualifying months, long enough for a full shock-and-recovery window. Detection, however, runs on any contiguous run longer than the baseline and shock requirement combined. That requirement is $6 + 2 = 8$ months, and the implementation admits runs *exceeding* it, so the shortest usable run is nine months: eight would leave no month in which recovery could be observed. A category split by sparse months contributes more than one such run. That yields 27 usable series across 46 categories. Series shorter than 26 months can host a shock but its outcome is necessarily censored, which accounts for part of the censoring rates below.

With that in hand, we ran detection on the 27 series under two sentiment measures: mean star rating, which requires no model, and lexicon polarity. Table 5 reports the results.

We read the null comparison from the star-rating measure alone. Under star ratings, 23 shocks were observed against 28.39 ± 3.77 expected under the block bootstrap null at 999 replicates, a ratio of 0.81 with empirical $p = 0.95$: *fewer* events than noise produces. Under lexicon polarity, 29 were observed against 26.19 ± 3.73 , a ratio of 1.11 with $p = 0.25$. These p -values are empirical quantiles of the 999-replicate bootstrap distribution under the $(b + 1)/(B + 1)$ convention. An earlier version of this paper reported

Table 5. Shock detection outcomes and null comparison, balanced panel at 75 % presence. The star-rating column carries the reported result: it is computed from the rating field alone and requires no extraction model and no annotation. The lexicon column is a robustness check on an unvalidated measure (Section 7) and is shown for comparison only; no conclusion in this paper rests on it.

	Star rating (primary)	Lexicon polarity (robustness only)
<i>Detection</i>		
Usable category series	27	27
Shock episodes	23	29
recovered	20	12
non-recovered	0	7
censored	3	10
Censoring rate	13.0 %	34.5 %
Analysable episodes	20	19
EPP (3 parameters)	6.7	6.3
EPP (6 parameters)	3.3	3.2
<i>Null comparison</i>		
Block bootstrap null (mean)	28.39	26.19
Block bootstrap null (SD)	3.77	3.73
Observed / block null	0.81	1.11
Empirical p	0.95	0.25
Ratio 95 % CI	[0.64, 1.15]	[0.83, 1.53]

normal approximations computed from 30 replicates; those values, $p = 0.91$ and $p = 0.25$, are close to the empirical ones but are superseded. Figure 7 places each observation within its null density.

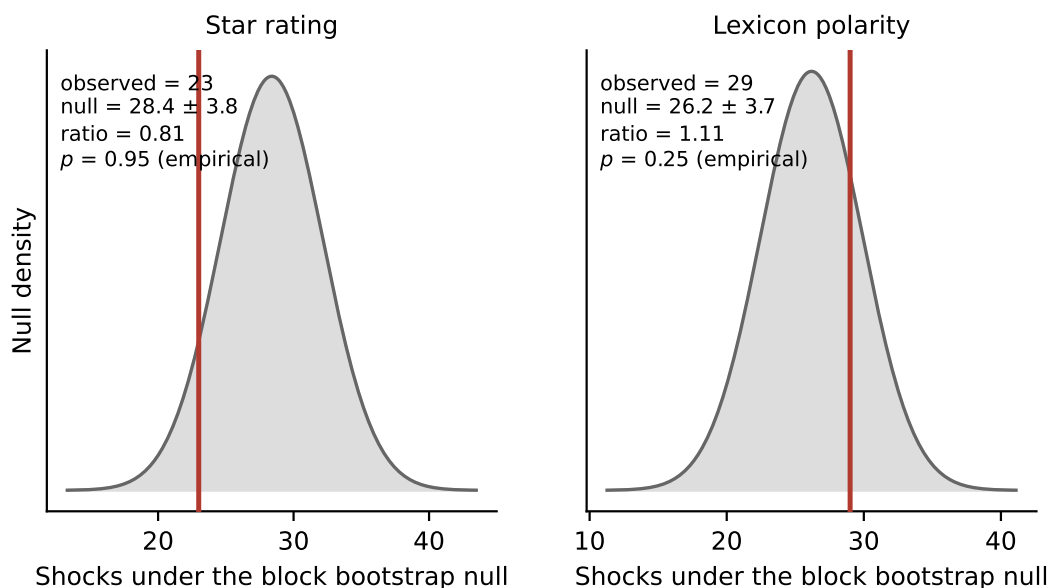


Figure 7. Observed shock counts (vertical lines) against the block bootstrap null density for each sentiment measure. Both observations fall inside the bulk of the null distribution; the star-rating count falls below the null mean.

5.6 Does the instrument work? A positive control

A null result is uninformative unless the detector is shown to detect. Every test written for this study checks classification logic: that a truncated decline is censored, that a gap splits a series. None measures whether the criterion recovers a shock that is actually present. We therefore ran a positive control, injecting falls of known magnitude and duration into AR(1) series matched to the observed panel in length, level and autocorrelation, and measuring how often the criterion recovers them.

The synthetic geometry is stated rather than claimed to be matched. Series are 40 months long, which is the modal length of the qualifying runs in Figure 6 but not their range, which spans nine to 43 months. The level is 4.0 on the five-point rating scale and the innovation standard deviation 0.10. Autocorrelation is set at $\phi = 0.6$, a moderate value we chose rather than estimated from the panel. Power is reported against ϕ in the repository output and falls from 0.89 at $\phi = 0$ to 0.59 at $\phi = 0.85$ for a two-standard-deviation shock, so the figure we report sits in the middle of that range rather than at its favourable end. The one quantity we can check against the data is the false positive rate, and it agrees with the observed bootstrap null; that agreement is the evidence that the synthetic series are not unrepresentative.

The result is not favourable to our instrument and we report it in full (Table 6).

Table 6. Detection power of the pre-specified criterion on synthetic AR(1) series: 27 series of 40 months at $\phi = 0.6$, one injected shock each, so each cell rests on 27 trials and carries a standard error of roughly 0.09. Differences of one or two cells are therefore not interpretable, and the non-monotonicity at 0.5 SD should be read as noise. Power is the share of injected shocks recovered within two months of their true onset. The false positive column counts clean series, with nothing injected, on which the criterion nonetheless fires.

Injected shock	2 months	3 months	6 months	False positives
0.5 SD	0.56	0.44	0.56	1.00
1.0 SD	0.56	0.48	0.67	1.00
1.5 SD	0.56	0.59	0.70	1.00
2.0 SD	0.63	0.67	0.74	1.00
3.0 SD	0.70	0.78	0.89	1.00

Two things follow, and the second matters more than the first.

At its own criterion, a fall of one standard deviation sustained two months, the rule recovers 56 % of injected shocks. At three standard deviations sustained six months it reaches 89 %. Power rises with shock size, as it must, but slowly: a sixfold increase in magnitude buys thirty percentage points. The instrument discriminates weakly.

The false positive rate is 1.00. Every clean series, containing no injected shock at all, produces at least one detection. This is not a defect discovered late; it is the quantitative form of what the block bootstrap null was already telling us. The null mean of 28.39 detections across 27 series is 1.05 per series, and the synthetic control gives 1.00 per series. The two agree, which is also a check that the synthetic series resemble the real ones.

The consequence for our claims is direct. We cannot say a shock would have been found had one existed, because at realistic magnitudes the criterion finds roughly two-thirds of them. And our observed count of 23 across 27 series, or 0.85 per series, is *below* what the criterion produces on noise. The null comparison is therefore correct in what it reports and cannot support the stronger reading that the data contain no events. A criterion with a unit false positive rate can only ever say that the data are not noisier than noise.

We report this rather than removing the criterion because it was pre-specified, because its failure mode is informative for anyone adopting a similar rule, and because the change point analysis that follows does not share it.

5.7 Change point detection finds what the threshold rule does not

The protocol named PELT as a robustness check on shock localisation. An earlier version of this paper argued the check was moot because detection had already failed against the null. That argument was wrong, and the run shows why: the null comparison counts events, PELT locates them, and the two questions have different answers.

On the same 27 category series under star ratings, PELT identified six change points against 1.77 ± 1.35 expected under the same block bootstrap null: a ratio of 3.40, $z = 3.15$, empirical $p = 0.015$. Under the threshold rule the same series gave a ratio of 0.81 with $p = 0.95$.

Table 7. Two detection procedures on identical series, each against the same circular block bootstrap null at 999 replicates. The disagreement is the result, not an inconsistency: the threshold rule asks how many sustained falls of a given magnitude occur, and PELT asks where the mean shifts.

	Star rating		Lexicon	
	Threshold	PELT	Threshold	PELT
Observed	23	6	29	1
Null mean	28.39	1.77	26.19	0.80
Null SD	3.77	1.35	3.73	0.86
Ratio	0.81	3.40	1.11	1.26
Empirical p	0.95	0.015	0.25	0.56

Three qualifications belong with this result and none of them is a reason to set it aside.

The lexicon measure does not reproduce it: one change point against 0.80 expected, $p = 0.56$. Two measures of the same series again disagree, as they did on outcome composition, and we cannot say which is right.

Six events across 27 series is a small number. The significance rests on the null being small too, and a null mean below two makes the normal approximation uncomfortable, which is why we report the empirical p .

The PELT penalty, $3 \log(n) \sigma^2$, is a convention we adopted rather than tuned, and change point counts are sensitive to it. A penalty chosen to produce this result would be indefensible; we did not choose it that way, but neither did we test alternatives, and a confirmatory study should.

A fourth qualification is decisive and we state it plainly. Four tests were run on the same 27 series against the same null: two procedures crossed with two sentiment measures. Under Bonferroni correction across that family, $p = 0.015$ becomes $p_{\text{adj}} = 0.060$, and the PELT result no longer reaches conventional significance. An earlier version of this paper asserted that multiple comparison correction "would only strengthen the reported conclusion", which was true when every test was null and ceased to be true the moment one was not. We have no principled argument for treating these four as separate families, since they share a dataset, a null and a research question.

What we take from the result is therefore narrower than a positive finding. Category-level star rating series show temporal structure that is suggestive against a block bootstrap null and does not survive correction for the number of tests performed. Our pre-specified event criterion does not capture that structure at all. We report the uncorrected and corrected values together and draw no conclusion from either.

5.8 Two findings, at two different levels

Before the aspect analysis we separate two results that are easy to merge and should not be.

The first is that whole-review sentiment at category level produces shock counts indistinguishable from noise. That is a statement about *category-level, whole-review* sentiment, measured on the star rating.

The second is that aspect-level series do not form at all. That is a statement about *density*, and it is logically prior: where no series exists, no null comparison is possible.

Our title concerns the second. The first is reported because it is what the data permitted and because it bears on the wider literature, not because it establishes anything about aspects. A reader should not take the null result as evidence about aspect-level recovery; the evidence about aspect-level recovery is that it cannot be attempted.

5.9 Aspect-level series do not form at all

The analyses above measure sentiment over whole reviews. That is category-level sentiment, and a study about attributes cannot rest on it. We therefore applied the aspect taxonomy to the same balanced panel and rebuilt the series as (category, aspect, month).

Splitting 74,780 reviews across 46 categories, 10 aspects and 48 months gives 22,080 possible cells, of which 7,084 were non-empty (32.1 %), at a median of one mention each. **None reached the 30-mention threshold, and no aspect series survived to be tested.** Figure 8 gives the breakdown: performance and design form the most cells but still carry a median of two mentions; delivery, the sparsest, forms 165 cells at a median of one.

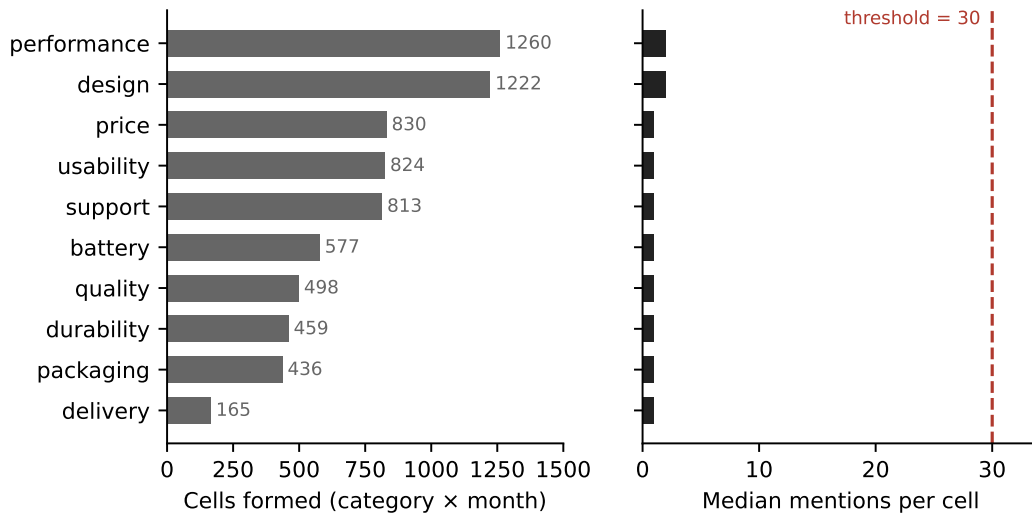


Figure 8. Aspect cells formed on the balanced panel (left) and their median mention count against the threshold (right). No aspect approaches it.

5.10 A bound independent of the extractor

Our lexicon was built for physical products and maps imperfectly onto apps: *delivery* and *packaging* mean little for software. Low coverage could therefore reflect a domain mismatch rather than sparsity, and that objection cannot be met by defending the lexicon.

It can be met by arithmetic. Let R be the mean reviews per unit-period, m the mean number of aspects a review mentions, and T the mention threshold. Total mentions available per unit-period is at most Rm , so the number of aspects j that can simultaneously reach T satisfies

$$j \leq \frac{Rm}{T}. \quad (2)$$

This does not assume even distribution. Skewed extraction concentrates mentions, which can lift one aspect above the threshold, but it cannot raise the total: whatever one aspect gains another loses. Equation 2 therefore bounds how many aspects can qualify under any assignment, uniform or not. It is an upper bound and not an estimate: the realised number may be lower and never higher, and under strong skew it can be zero even where the bound permits one.

For the balanced MobileRec panel, $R = 74,780/2,150 = 34.8$ and $T = 30$. Even at $m = 1$ the bound gives $j \leq 1.16$, so at most one aspect of any taxonomy can qualify; at $m = 3$, at most three. An earlier

version of this paper used $R = 33.9$, dividing by the $46 \times 48 = 2,208$ maximum rather than the 2,150 category-months the panel contains. The logged denominator is the correct divisor and the figures here use it. A ten-aspect study is infeasible unless reviews mention nine aspects each. Under the even-distribution case the shortfall is starker still: expected mentions per cell is $34.8/k$, meeting the threshold only at $k = 1$, falling to 17.4 at $k = 2$ and 3.5 at $k = 10$ (Figure 9).

The Amazon extract fails the bound more decisively. Its 82,041 reviews span 11,925 product-months, or $R = 6.9$ reviews per product-month. At $m = 1$ the bound gives $j \leq 0.23$: not one aspect can qualify, and the shortfall is a factor of 4.4 before any aspect split is applied at all. This is why no aspect-level analysis was attempted on Amazon. It would have been arithmetic performed on an empty set.

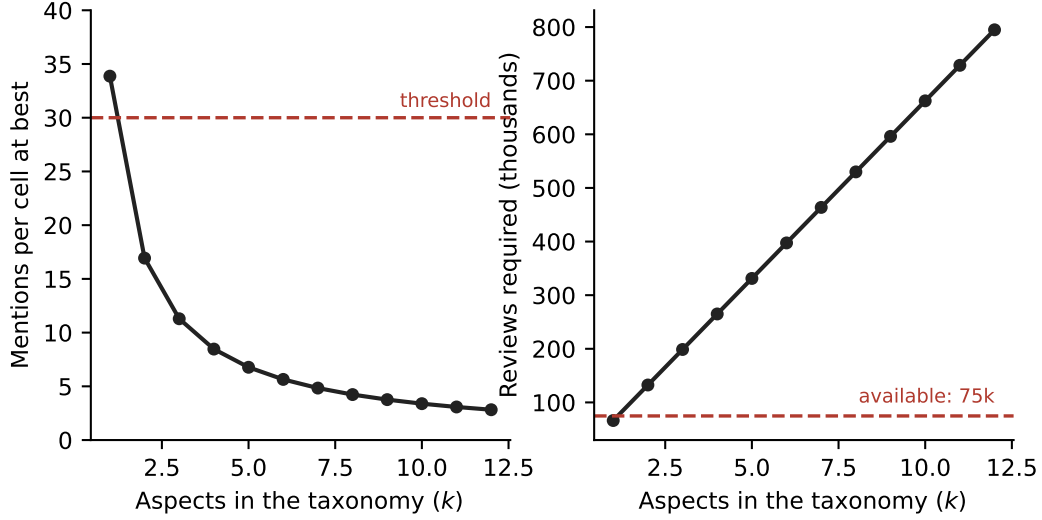


Figure 9. Best-case mentions per cell (left) and reviews required for full coverage (right) against taxonomy size k . Required reviews are $30k \times 46 \times 48$ at $m = 1$; at $k = 10$ this is 662,400. The reference line marks the 74,780 reviews the balanced panel holds. The bound assumes an extractor that never misses and never over-assigns, so no real extractor can do better.

Two parameters govern the bound and neither was measured. We report both rather than fixing them silently.

The first is m . We never estimated the mean number of aspects a review names, because our lexicon was not validated and an unvalidated extractor cannot estimate it. Every statement above is therefore conditional on m , and where we write a single number we have set $m = 1$. That is the conservative choice for our argument only in the sense that it is the smallest defensible value; it is not a measurement, and a reader should treat $j \leq 1.16m$ rather than $j \leq 1.16$ as the result.

The second is T . Table 8 gives the bound across thresholds. The conclusion is not threshold-free: at $T = 30$ the MobileRec panel supports at most $1.16m$ aspects, but at $T = 10$ it supports $3.48m$, so a ten-aspect taxonomy becomes arithmetically possible once $m \geq 3$. What the table shows is the combination required, and we make no claim beyond it.

Table 8. The density bound $j \leq Rm/T$ across mention thresholds, for the MobileRec balanced panel ($R = 34.8$) and the Amazon extract ($R = 6.9$). Entries are coefficients on m : the maximum number of aspects that can reach the threshold is the entry multiplied by the mean aspects named per review. Amazon fails at every threshold we would consider defensible.

Threshold T	5	10	15	20	30	50
MobileRec, category	$6.96m$	$3.48m$	$2.32m$	$1.74m$	$1.16m$	$0.70m$
Amazon, product	$1.38m$	$0.69m$	$0.46m$	$0.35m$	$0.23m$	$0.14m$

A low threshold is not free. At $T = 5$ a monthly cell mean rests on five mentions, and the shock criterion compares such a mean against a standard deviation estimated from six of them. We fixed $T = 30$

before seeing data for that reason, and we do not revisit it now; but the honest statement is that our negative conclusion holds at the threshold we pre-specified and weakens as the threshold falls.

The bound holds whatever the extraction method, because no extractor can recover an attribute judgement from a review that was never written. An aspect-level study of recovery on this corpus is not limited by method but by how much text exists.

This is the appropriate answer to the natural objection that a transformer or instruction-tuned model would extract more than a lexicon. It would, and we do not dispute it. A better extractor raises m , the mean aspects recovered per review, and Equation 2 is linear in m . Reaching ten qualifying aspects on the MobileRec panel would require $m \geq 8.6$: every review would have to yield judgements on close to nine distinct attributes. Reaching even three would require $m \geq 2.6$. These are not extraction targets; they are statements about how much is said in a typical app review. We did not run a transformer baseline for this reason, and we state the omission plainly rather than implying it was unnecessary: a reviewer who believes $m \geq 3$ is attainable on short app reviews should regard this part of our argument as untested.

5.11 Time to recovery

With the episodes classified, a Kaplan–Meier estimate of time to recovery is available and we report it rather than describing why we did not.

Under star ratings the median time to recovery is three months, with the survival curve falling to zero by month 14 and no episode reaching the 18-month horizon uncovered. Under lexicon polarity the median is 15 months and the curve does not reach zero. A five-fold difference between two measures of the same episodes is not a result about recovery; it is further evidence that the episodes are not stable objects.

Confidence bands are wide throughout, as 23 and 29 episodes require. We fit no Cox model: at 20 and 19 analysable episodes the events-per-parameter figures in Table 5 rule it out, and a hazard ratio estimated from them would carry a precision the data cannot support.

5.12 What the null comparison could have detected

A null comparison that fails to reject is uninformative unless its own sensitivity is stated. Treating the bootstrap null as approximately normal, the one-sided critical value at $\alpha = 0.05$ is 33.7 shocks for star ratings and 33.2 for lexicon polarity, ratios of 1.21 and 1.28. At 80 % power the smallest detectable ratios are 1.32 and 1.42.

Our result therefore rules out a signal inflating shock counts by roughly a third or more above chance. It does not rule out a smaller one, and we claim no such thing. Detecting a 10 % excess at this null variance would need about an order of magnitude more series than the 27 available, which returns the problem to the density constraint of Section 5.

5.13 The two measures disagree

Star ratings produced no non-recovered episode, while lexicon polarity produced seven. An earlier version of this paper offered the divergence as evidence that the episodes were not real events. That inference does not hold. A mean star rating is an overall judgement and a lexicon score is a property of review text; they are not two measurements of one construct, and divergence between them is expected rather than diagnostic. We report the difference as a description of the two measures and draw no inference from it.

The comparison retains one use. The star-rating measure requires no model and no annotation, so a reader who distrusts the unvalidated lexicon can read every reported conclusion from the star-rating column alone. Nothing in this paper depends on the lexicon.

Figure 10 shows the outcome composition under each measure. We display it because the two columns of Table 5 invite the comparison, not because the difference supports an inference; for the reason given above, it does not.

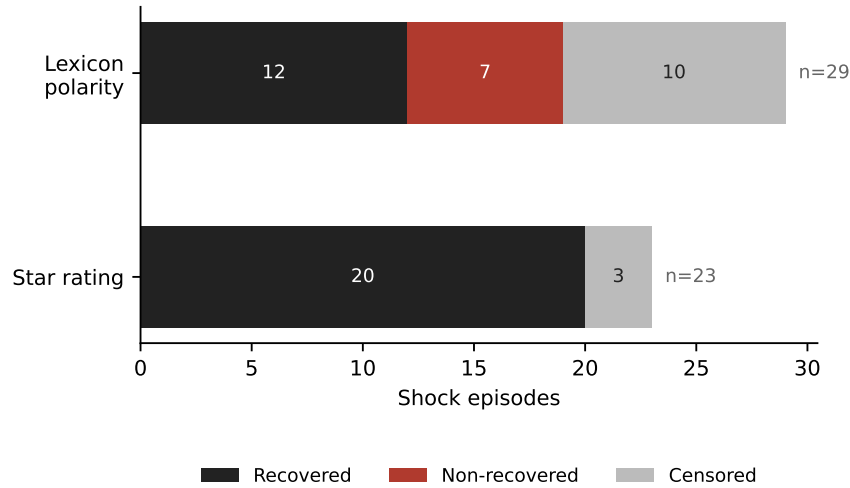


Figure 10. Outcome composition on identical series: 23 episodes under star rating, 29 under lexicon polarity. Star ratings produced no non-recovered episode; see Section 5.13 for why that is not interpretable on its own.

5.14 Sensitivity

Figure 11 reports shock counts across the detection parameter grid. Counts fall monotonically as either the magnitude threshold or the duration requirement is tightened, as expected, and no cell yields a count that would change the power conclusion. We note that relaxing thresholds until events appear would be the natural response to these results and would be invalid: events recovered that way are the noise the null comparison already identifies.

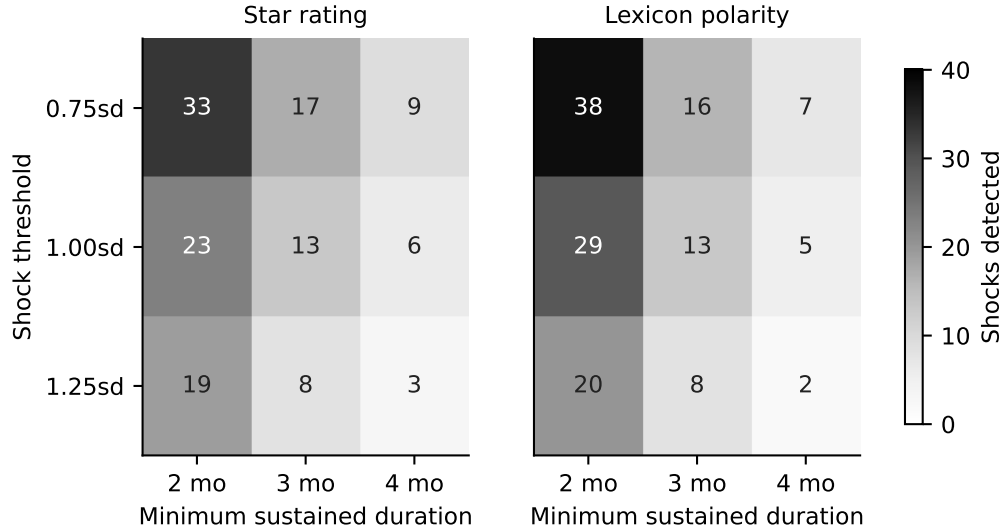


Figure 11. Shocks detected across the detection parameter grid for both measures. Counts fall monotonically as either criterion tightens; no cell yields a count that would change the power conclusion, and none was significant against the null.

5.15 Methodological controls and what each prevents

Table 9 lists the controls and the specific failure mode each addresses. Seven were implemented and exercised, each with an automated test asserting its behaviour on a synthetic case with a known correct answer. The last of them, the positive control of Section 5.6, was added after an earlier version of this paper was reviewed and is the one that most changed what we could claim. One control was specified

in the protocol but never exercised, because the analysis it guards was never reached; they are marked accordingly, and we list them because the failure modes they describe apply to any study that does reach those analyses.

Table 9. Controls implemented, the artefact each prevents, and the direction of the bias if the control is omitted.

Control	Failure mode prevented	Direction of omitted-control bias
<i>Implemented and exercised</i>		
Right-censoring of short follow-up	Shocks near the end of the window cannot be observed to recover	Overstates irreversibility; grows with the horizon
Circular block bootstrap null	Permutation removes autocorrelation and raises the detection threshold	Understates the null; favours detection
Cohort vs population recovery	A returning average may reflect new reviewers, not changed minds	Attributes turnover to perceptual recovery
Gap-aware segmentation	Series with dropped months are otherwise treated as contiguous	Creates spurious shocks across discontinuities
Balanced panel	Category membership recomputed each month conflates composition with level	Attributes composition change to sentiment change
Guarded ratio reporting	Recovery attribution is unstable when the denominator approaches zero	Emits plausible but meaningless ratios
Positive control	A detector may fail silently on shocks that are present	Would let a null result be read as evidence of absence
<i>Specified but never exercised</i>		
Time-varying recovery covariate	Recovery status requires survival to be observed, creating immortal time	Manufactures a protective effect of recovery

5.16 A note on dataset provenance

The Amazon Reviews 2023 collection pipeline is user-centric: user identifiers are sampled first, and all review pages associated with each sampled user are then retrieved. The authors note that item-level review coverage may be incomplete when some reviewers of an item fall outside the user pool (Hou et al., 2026). This is appropriate for the recommendation tasks the dataset targets, but it means item-level series are thinned by construction, independently of how often customers review. Our product-level density figures should therefore be read as a property of this corpus as well as of reviewing behaviour. Both corpora we examined are sampled user-centrally, and we did not survey others. We make no claim about what public corpora exist: a systematic review of candidates such as Yelp, Goodreads, Steam or TripAdvisor against the requirements in Section 6.3 would be a useful exercise and we have not performed it. What we can say is that neither corpus we used provides item-complete longitudinal coverage, and that an item-centrally sampled corpus would remove one of the two constraints described above, leaving the behavioural one intact.

Table 10 sets out the properties each corpus offers against what the design requires.

Table 10. Design requirements against corpus properties. A requirement is met only if it holds at the unit of analysis the design uses. *n.t.*: not tested, because the configuration failed an earlier requirement and the step was never reached. For Amazon specifically, no aspect extraction was attempted because the density bound of Section 5.10 rules out any aspect reaching the threshold at $R = 6.9$, before extraction quality enters.

Requirement	Amazon 2023	MobileRec (app)	MobileRec (category)
Timestamped reviews	yes	yes	yes
Persistent user identifier	yes	yes	yes
≥ 30 mentions per unit-month	2.6 %	0.2 %	44.4 %
≥ 26 contiguous qualifying months	no	no	16 categories
Repeat review of the same item	0.16 %	0.06 %	0.06 %
Aspect cells above threshold	n.t.	n.t.	0 of 7,084
Signal above block bootstrap null	n.t.	n.t.	no
Documented intervention dates	no	no	no

6 Discussion

6.1 The obstacle is behavioural

Three configurations failed for three different proximate reasons: density, repeat observation, and signal-to-noise. Two causes compound. Most reviewers write once and do not return, which is a property of behaviour. And the corpora available at this scale are assembled user-centrally, which thins item-level series further (Hou et al., 2026). The first cause cannot be engineered away by choosing a different corpus; the second could be, but we are not aware of a public corpus that does so at the required scale. Choosing a third corpus with the same behavioural profile and the same sampling design would reproduce the result; an item-centrally sampled corpus would relieve the second constraint but not the first.

6.2 A test others can apply before committing

The bound in Section 5.10 is arithmetic and transfers. Before designing an aspect-level temporal study, divide the reviews available in the intended panel by $(\text{units} \times \text{periods} \times \text{aspects})$ and compare against whatever mention threshold the analysis needs. If the quotient falls short, no choice of extractor or model will rescue the design, and the remedy is more text or coarser time periods rather than a better method. We claim nothing for the inequality itself, which is elementary. We claim only that computing it before assembling a panel is not standard practice and that stating it in a form that transfers has some use as a first step in study design. We did not compute it first, which is why this paper exists.

6.3 What data would make this estimable

Inverting Equation 2 turns the negative result into a specification. For a study of j aspects at threshold T over P periods and U units, the corpus must supply at least $UPjT/m$ reviews, and they must be distributed across units and periods rather than concentrated at launch. For a modest design of five aspects, monthly periods over three years, fifty units and $T = 30$ at $m = 2$, that is 135,000 reviews *within the panel*, or 1.8 times the balanced panel we assembled from a corpus of 19.3 million interactions.

Three further properties are needed and none of them follows from volume. Repeat observation of the same user on the same item, at a rate high enough to estimate a cohort, is required to separate perceptual recovery from turnover; neither corpus here comes within two orders of magnitude. Item-centric rather than user-centric sampling is required so that item series are not thinned by construction. Item-centric sampling matters in both corpora studied here, not only in Amazon: MobileRec’s 5-core filter is applied to users, so items reviewed by few sampled users are thinned in the same way. And documented intervention dates are required if recovery is to be attributed to anything: without them the analysis can describe when sentiment moved but never why.

We suspect such data exists inside firms and not outside them. That is a limitation of public corpora rather than of the construct, and it suggests that progress on recovery dynamics will come from industry collaboration or from purpose-built collection, not from re-mining the corpora already available.

6.4 Implications for recovery claims

Our findings do not show that sentiment recovery does not occur. They show that on public review data it is not identifiable in the way the construct requires. Two consequences follow for work that reports recovery-like patterns.

First, in the absence of repeat individual observation, a returning average is consistent with perceptual recovery and with population turnover, and these cannot be told apart. Reporting the former without excluding the latter overstates what the data support.

Second, results at aggregated levels require an explicit null. Our category series were dense enough to produce plausible shock counts with plausible recovery patterns; only the block bootstrap comparison revealed that these were not distinguishable from noise. A study reporting Table 5 without its lower panel would appear to have found something.

6.5 Null choice is not a detail

As shown in Section 3.4, the permutation null is biased toward the alternative because it removes the autocorrelation that sets the detection threshold. Under star ratings the permutation null returned 25.8 against the block bootstrap’s 27.8; the direction of the bias matches our synthetic demonstration. Studies of temporal review dynamics that validate against permutation nulls, or against no null at all, may be reporting detection rates that a suitable control would not support.

7 Limitations

Our search of the prior literature was targeted rather than systematic, so we cannot claim the gap identified in Section 2.4 is unoccupied; a search conducted and reported under PRISMA (Page et al., 2021; Moher et al., 2009) would be required. Our analyses use subsamples of both corpora rather than their full extent, and a larger extract could change the density figures, though it would not change the repeat-review rate, which is a property of behaviour rather than sample size. The lexicon used for polarity is deliberately crude, and we must be explicit about what was not done. Its weakness bears on Section 5.9 but not on Section 5.10, whose bound is independent of the extractor. Our protocol specified validation of extraction against 300 human-labelled reviews, with agreement reported as Cohen’s κ . That validation was never carried out, because the feasibility constraints in Section 5 made the downstream analysis moot before annotation was reached. The lexicon measure is therefore unvalidated and any inference resting on it alone should be discounted accordingly; we report it only alongside the star-rating measure, which requires no model and no annotation, and which produced the weaker of the two null ratios. A trained aspect-based model might detect structure our measure misses, and that possibility is not excluded here.

Four tests were run across two procedures and two sentiment measures on one set of series. Bonferroni correction across that family is applied in Section 5.7 and changes the one suggestive result from $p = 0.015$ to $p = 0.060$. We do not additionally correct across the 27 individual series, since no per-series result is reported. Finally, the category-level analysis was exploratory: it was not covered by the protocol, and its results are reported as exploratory.

We did not screen for coordinated review campaigns, which produce shock-like signatures and are a known contaminant of review corpora. Opinion spam was characterised by Jindal and Liu (2008) and Ott et al. (2011), group manipulation by Mukherjee et al. (2012), and detection methods are surveyed by Mohawesh et al. (2021) and Paul and Nikolaev (2021). Manipulation has measurable commercial value (Luca, 2011), which is why it persists. Because our conclusion is that detected shocks are indistinguishable from noise, undetected manipulation would if anything work against that conclusion by adding real structure; screening would nevertheless be required in any confirmatory analysis.

Two analyses specified in the protocol remain unrun. No Cox model was fitted, because 20 analysable episodes fall an order of magnitude below what one needs; Kaplan–Meier is reported instead (Section 5.11). And no transformer or instruction-tuned extractor was compared against the lexicon, which matters more than it did in earlier versions of this paper because the density bound is stated in terms of m and m is unmeasured. A reader who believes a modern extractor would raise m substantially on short app reviews should treat Section 5.10 as untested at that point.

Two others were unrun when an earlier version of this paper was written and have since been carried out. The bootstrap now uses 999 replicates with empirical p -values rather than 30 with a normal approximation. PELT was run, and its result disagrees with the threshold rule (Section 5.7); we had previously argued the check was unnecessary, and that argument was wrong.

We also note that a category-level shock is not the same phenomenon as a product-level one. It averages hundreds of apps and may reflect store ranking changes, operating system releases, or seasonal effects rather than perceptual change. We did not attempt to separate these, because the null comparison made the question moot.

8 Conclusion

We set out to model attribute-level sentiment decline and return as a time-to-event process and found that the subsamples examined here cannot support it. Two results, at two levels, should not be run together.

At the aspect level the series do not form: none of 7,084 cells reached the mention threshold. The density bound explains why and says how far the shortfall extends, at $j \leq 1.16m$ for the MobileRec panel and $0.23m$ for the Amazon extract at $T = 30$. Because m was never measured, this is a conditional statement, and Table 8 shows how it moves with the threshold.

At category level, where whole-review sentiment does yield series, the result is mixed and we resist resolving it. Our pre-specified event criterion found shock counts indistinguishable from a block bootstrap null, at a ratio of 0.81. PELT, applied to the same series against the same null, found change points at a ratio of 3.40, $p = 0.015$ uncorrected and $p = 0.060$ under Bonferroni across the four tests performed. Temporal structure is present in category-level star ratings; our criterion does not recover it; and the lexicon measure reproduces neither result. We report all three rather than selecting the one that reads most cleanly.

This bears on the paper’s scope. We can say that an aspect-level time-to-event study is not possible on these corpora, and the density bound explains why in terms that transfer. We cannot say that category-level review series are structureless, and the PELT result is evidence that they are not. At the product and app levels the series are one to two orders of magnitude too sparse and the resulting event counts fall short of the required power by more than an order of magnitude. Repeat reviewing of the same item is rare enough that recovery of a population average cannot be separated from turnover in the reviewing population. Aggregation to the category level restores density but yields shock counts indistinguishable from a block bootstrap null under either of two sentiment measures, which themselves disagree about outcomes.

We report this because the design is a natural one that others are likely to attempt, because feasibility failures are systematically under-reported (Franco et al., 2014), and because the failure points are informative independently of our particular hypotheses. The pre-specified definitions, the code, and the self-test suite are available so that the assessment can be checked or extended to other corpora.

Data and code availability

All code, pre-specified definitions, self-tests, and figure generation scripts: <https://github.com/ssvakil/aspect-recoverability>. Both datasets analysed are publicly available from their original distributors.

Author contributions

S. Vakil: conceptualisation, methodology, software, formal analysis, data curation, writing — original draft, visualisation. **Y. Farjami:** supervision, methodology, writing — review and editing.

Declaration of competing interests

The authors declare no competing interests.

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